NARAYANA COACHING CENTER

For Students of Class XII/XII PASS

NARAYANA ALL INDIA TEST SERIES

THE GATEWAY TO SUCCEED

for JEE Main & JEE Advanced, 2024

44 YEARS OF EXCELLENCE

Sec: NAITS **Date: 04-02-2024**

Time: 3 Hrs. **PTA – 01** **Max. Marks: 300**

JEE-ADV-2018-P1 MODEL

Time: 3:00Hour's **IMPORTANT INSTRUCTIONS** **Max Marks: 180**

PHYSICS:

Section	Question Type	+Ve Marks	- Ve Marks	No.of Qs	Total marks
Sec – I(Q.N : 1 – 6)	Questions with Multiple Correct Choice (partial marking scheme) (+1,0)	+4	-2	6	24
Sec – II(Q.N : 7 – 14)	Questions with Numerical Value Type (e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30)	+3	0	8	24
Sec – III(Q.N : 15-18)	Questions with Comprehension Type (2 Comprehensions – 2 +2 = 4Q)	+3	-1	4	12
Total				18	60

CHEMISTRY:

Section	Question Type	+Ve Marks	- Ve Marks	No.of Qs	Total marks
Sec – I(Q.N :19 – 24)	Questions with Multiple Correct Choice (partial marking scheme) (+1,0)	+4	-2	6	24
Sec – II(Q.N : 25 –32)	Questions with Numerical Value Type (e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30)	+3	0	8	24
Sec – III(Q.N : 33-36)	Questions with Comprehension Type (2 Comprehensions – 2 +2 = 4Q)	+3	-1	4	12
Total				18	60

MATHEMATICS:

Section	Question Type	+Ve Marks	- Ve Marks	No.of Qs	Total marks
Sec – I(Q.N:37 – 42)	Questions with Multiple Correct Choice (partial marking scheme) (+1,0)	+4	-2	6	24
Sec – II(Q.N :43–50)	Questions with Numerical Value Type (e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30)	+3	0	8	24
Sec –III(Q.N : 51-54)	Questions with Comprehension Type (2 Comprehensions – 2 +2 = 4Q)	+3	-1	4	12
Total				18	60

Exam Syllabus	
PHYSICS	Units and Measurement, Motion in a Straight Line,Motion in a Plane, Laws of Motion, Work, Energy and Power, System of Particles and Rotational Motion
CHEMISTRY	Redox reaction, Some Basic concept in Chemistry, Atomic Structure, Periodic Properties Chemical bonding and molecular structure States of Matter Thermodynamics
MATHEMATICS	Sequence and Series Quadratic Equations Binomial theorem Permutation and Combination Trigonometric functions Trigonometric Equations

Subject	Physics	Chemistry	Mathematics
Setter Branch	AURANGABAD		
SME Name	Mr. ABDUL WAHAB	Mr. D. RAO	MR. PRADEEP MISARA
Phone Number	93404 77855	99874 27722	93736 90662

Rough Work

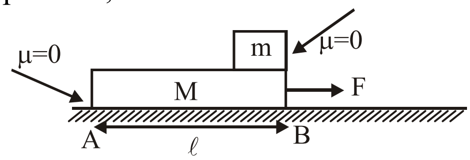
PHYSICS **MAX.MARKS: 60**

SECTION- I
(Maximum Marks : 24)

This section contains SIX (06) questions.
Each question has FOUR options for correct answer(s). ONE OR MORE THAN ONE of these four option(s) is (are) correct option(s).
For each question, choose the correct option(s) to answer the question.
Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If only (all) the correct option(s) is (are) chosen.
Partial Marks: +3 If all the four options are correct but ONLY three options are chosen.
Partial Marks: +2 If three or more options are correct but ONLY two options are chosen, both of which are correct options.
Partial Marks : +1 If two or more options are correct but ONLY one option is chosen and it is a correct option.
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).
Negative Marks: -2 In all other cases.

1. In the figure small block is kept on M, then



A) the acceleration of m w.r.t. ground is $\frac{F}{m}$

B) the acceleration of m w.r.t. ground is zero

C) the time taken by m to separate from M is $\sqrt{\frac{2\ell m}{F}}$

D) the time taken by m to separate from M is $\sqrt{\frac{2\ell M}{F}}$

2. A particle of mass m, moving with velocity v collides a stationary particle of mass $2m$. As a result of the collision, the particle of mass m deviates by 45^0 and has final speed of $\frac{v}{2}$. For this situation mark out the correct statement(s).

A) The angle of divergence between particles after collision is $\frac{\pi}{2}$

B) The angle of divergence between particles after collision is less than $\frac{\pi}{2}$

C) Collision is elastic

D) Collision is inelastic

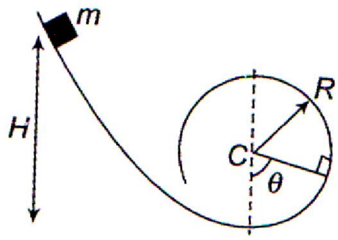
Rough Work

3. A particle of mass m moves along a curve y = x^2. When particle has x- co-ordinate as ½ and x-component of velocity as 4m/s then:
A) The Position coordinate of particle are (1/2, 1/4)
B) The Velocity of particle will be along the line 4x-4y-1=0.
C) The magnitude of velocity at that instant is $4\sqrt{2}$ m/s
D) The magnitude of angular momentum of particle about origin at that position is 0.

4. A man is standing on a road and observes that rain is falling at angle 45^0 with the vertical. The man starts running on the road with constant acceleration 0.5 m/s^2. After a certain time from the start of the motion, it appears to him that rain is still falling at angle 45^0 with the vertical, with speed $2\sqrt{2}$ m/s. Motion of the man is in the same vertical plane in which the rain is falling. Then which of the following statement(s) are true:
A) It is not possible
B) Speed of the rain relative to the ground is 2m/s.
C) Speed of the man when he finds rain to be falling at angle 45^0 with the vertical, is 4m/s.
D) The man has travelled a distance 16m on the road by the time he again finds rain to be falling at angle 45^0

5. The gas equation for n moles of a real gas is $\left(P+\frac{a}{V^2}\right)(V-b)=nRT$ where P is the pressure, V is the volume, T is the absolute temperature, R is the molar gas constant and a, b are arbitrary constants. Which of the following have the same dimensions as those of PV?
A) nRT B) a/V C) Pb D) ab/V^2

6. A particle of mass m is released from a height H on a smooth curved surface which ends into a vertical loop of radius R. as shown in figure. Choose the correct alternatives(s) if H = 2R



A) The particles reaches the top of the loop with zero velocity
B) The particle cannot reach the top of the loop
C) The particle breaks off at a height h = R from the base of the loop
D) The particle breaks off at a height h from base of loop such that R < h < 2R.

Rough Work

SECTION - II
(Maximum Marks : 24)

This section contains **EIGHT (08)** questions. The answer to each question is a **NUMERICAL VALUE**

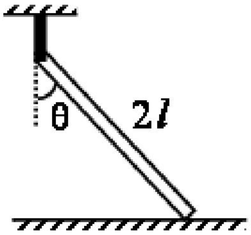
For each question, enter the correct numerical value (in decimal notation, truncated/rounded off to the **second decimal place**; e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30) designated to enter the answer.

Answer to each question will be evaluated according to the following marking scheme:

Full Marks: +3 If ONLY the correct numerical value is entered as answer.

Zero Marks: 0 In all other cases.

7. A uniform rod of length $2l$ and mass m is hanging to a string and touching the smooth floor as in the figure. The tension in the string when $\theta = 30°$ is T. The tension in the string when $\theta = 60°$ is $\frac{T}{n}$. Find n .



8. A 3.0 kg block slides on a frictionless horizontal surface, first moving to the left at 50 m/s. It collides with a non –ideal spring as it moves left, compresses the spring and is brought to rest momentarily. The body continues to be accelerated to the right by the force of the compressed spring. Finally, the body moves to the right at 40 m/s. The block remains in contact with the spring for 0.020s. The magnitude of the impulse of the spring on the block is $30n$ then n =________ .

9. A particle of mass 10^{-2} kg is moving along the negative x-axis under the influence of a force $F(x) = \frac{-K}{2x^2}$ where k=10^{-2} Nm^2. At time t=0 it is at x=1.0 m and its velocity is v=0 Find its velocity when its reaches x=0.50 m

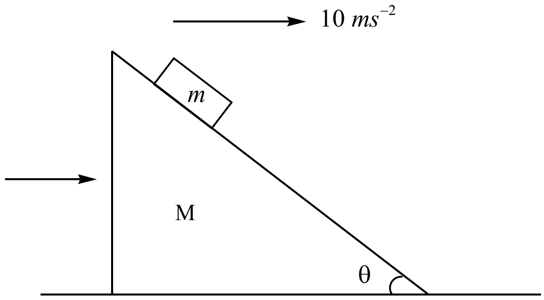
10. A point moves along a circle having a radius 20cm with a constant tangential acceleration $5cms^{-2}$. How much time is needed after motion begins for the normal acceleration of the point to be equal to tangential acceleration?

11. A rod is given an angular acceleration α from rest so that it rotates in horizontal plane about a vertical axis. It has a ring at a distance r from the axis of rotation. The friction coefficient between the ring and the rod is μ. Neglecting gravity. The time after which the ring will start to slip on the rod is $\frac{1}{n}$ sec. Then find the value of n (Take $\alpha = 3 rad\, s^{-2}$ and $\mu = 1/3$)

Rough Work

12. A body projected vertically up crosses a point P in its path at the end of 2 second and the highest point at the end of 5 second. After how many second from the start will it reach P again

13. In the figure shown all the surfaces are frictionless, and mass of the block is $m = 100g$. The block and the wedge are held initially at rest. Now the wedge is given a horizontal acceleration of $10\ ms^{-2}$ by applying a force on the wedge, so that the block does not slip on the wedge. Then find the work done in joules by the normal force in ground frame on the block in 1 s.



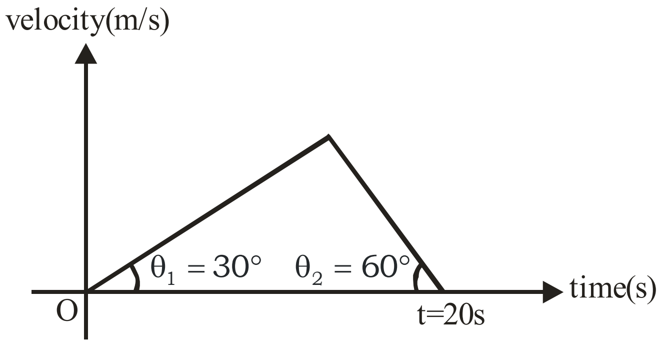
14. An object of mass 5 kg falls from rest through a vertical distance of 20 m and reaches a velocity of 10 m/s. How much work is done by the push of the air on the object? ($g = 10\ m/s^2$). (in J)

SECTION – III
(COMPREHENSIN TYPE)

This section contains 2Paragraphs. Based on each paragraph, there are 2 questions. Each question has 4 options (A), (B), (C) and (D) for its answer, out of which **ONLY ONE** option can be correct.
Marking scheme: +3 for correct answer, 0 if not attempted and -1 in all other cases.

Paragraph for Question Nos. 15 to 16

The velocity-time graph for a body on a straight path is shown in the figure.



Rough Work

15. The maximum speed of the body is

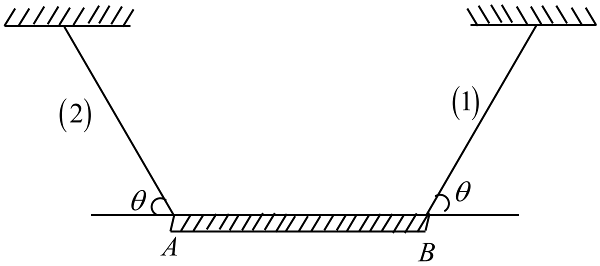
A) $\sqrt{3}$ m/s B) $\sqrt{5}$ m/s C) $5\sqrt{3}$ m/s D) $3\sqrt{5}$ m/s

16. The accelerating time of the body is

A) 5s B) 7.5s C) 10s D) 15s

Paragraph for Question Nos. 17 to 18

A uniform rod AB of length 'L' mass M is suspended by two massless identical threads. Initially the rod AB is in equilibrium in the vertical plane as shown in fig.



17. If thread (1) suddenly breaks then the tension present in other thread (2) is

A) $\dfrac{Mg}{4\sin\theta}$ B) $\dfrac{Mg}{4(1+\theta)}$ C) $\dfrac{Mg\sin\theta}{1+3\sin^2\theta}$ D) $\dfrac{Mg}{2(2+\cos\theta)}$

18. If the thread(1) suddenly breaks then the angular acceleration of the rod about it's centre of mass is

A) $\dfrac{3g}{2L\sin\theta}$ B) $\dfrac{3g}{2L(1+\cos\theta)}$ C) $\dfrac{6g\sin^2\theta}{L(1+3\sin^2\theta)}$ D) $\dfrac{6g\sin\theta}{L(1+3\sin^2\theta)}$

Rough Work

CHEMISTRY MAX.MARKS: 60

SECTION- I
(Maximum Marks : 24)

This section contains SIX (06) questions.
Each question has FOUR options for correct answer(s). ONE OR MORE THAN ONE of these four option(s) is (are) correct option(s).
For each question, choose the correct option(s) to answer the question.
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Partial Marks: +2 If three or more options are correct but ONLY two options are chosen, both of which are correct options.
Partial Marks : +1 If two or more options are correct but ONLY one option is chosen and it is a correct option.
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).
Negative Marks: -2 In all other cases.

19. Which of the following statements are correct for a given element ?
(A) The total spectral lines obtained from a single line during Zeeman effect is $(2l + 1)$.
(B) In the Lyman series as the energy liberated during transition increases then the distance between the spectral lines goes on decreasing.
(C) The highest probability of finding an electron in 1s orbital is in the vicinity (region around) of the circumference.
(D) The highest probability of finding an electron in 1s orbital is exactly at the middle between nucleus and circumference.

20. Which of the following compound or ion is/are having pyramidal shape
(A) ClO_3^- (B) $(SiH_3)_3N$ (C) $(CH_3)_3N$ (D) $\overset{\oplus}{C}H_3$

21. Which of the following statements is/are correct?
(A) The second IE of boron is greater than carbon.
(B) First IE of S is greater than first IE of P.
(C) The second IE of Mg is greater than that of P.
(D) First IE of Al and Ga are Almost same.

22. 180 gm of water is evaporated slowly under isothermal & isobaric conditions at 100ºC. Assuming water vapour to behave ideally ($\Delta H_v = 2260$ J/g) -
(A) $\Delta U = 375.8$ kJ (B) $\Delta S = 177.5$ J/K (C) $\Delta G = 0$ (D) $Q = 406.8$ kJ

Rough Work

23. On amu scale, atomic weight of Ca is 40. If one u is redefined, which of the following would be correct?

(A) Atomic weight of Ca is 20 if one u is defined as $1/6^{th}$ of weight of an atom of C-12

(B) Atomic weight of Ca is 60 if one u is defined as $1/15^{th}$ of weight of an atom of C-12

(C) Atomic weight of Ca is 70 if one u is defined as $1/21^{st}$ of weight of an atom of C-12

(D) Atomic weight of Ca is 80 if one u is defined as $1/30^{th}$ by weight of an atom of C-12

24. For complete oxidation of FeC_2O_4 it requires.

(A) 3/5 mole of acidified $KMnO_4$ solution (B) $\frac{1}{2}$ mole of acidified $K_2Cr_2O_7$ solution

(C) 2 mole of HNO_3 (D) 2 mole of Sn^{2+} solution.

SECTION - II
(Maximum Marks : 24)

This section contains **EIGHT (08)** questions. The answer to each question is a **NUMERICAL VALUE**

For each question, enter the correct numerical value (in decimal notation, truncated/rounded off to the **second decimal place**; e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30) designated to enter the answer.

Answer to each question will be evaluated according to the following marking scheme:

Full Marks: +3 If ONLY the correct numerical value is entered as answer.

Zero Marks: 0 In all other cases.

25. Consider the following table $n_1 to\ n_4$ are number of surrounding atoms.

Neutral Compounds (X are monovalent surrounding atoms)		Central atoms (A to D) belong to group	Characteristics of compounds	Number of lone pair(s) at central atom
(i)	AX_{n_1}	16	Planar and polar	m_1
(ii)	BX_{n_2}	15	Trigonal pyramidal	m_2
(iii)	CX_{n_3}	14	Zero dipole moment	m_3
(iv)	DX_{n_4}	13	All $X-\hat{D}-X$ bond angle are $120°$	m_4

Rough Work

Then calculate value of expression $\left|\dfrac{n_1+n_2+n_3+n_4}{m_1+m_2+m_3+m_4}\right|^2$

26. The enthalpy of vaporization and enthalpy of fusion of water are 546.5 cal/g and 80 cal/g. The ratio of $\Delta S_{Vap}/\Delta S_{fusion}$ for water is

27. At identical temperature and pressure, the rate of diffusion of hydrogen gas is $3\sqrt{3}$ times that of a hydrocarbon having molecular formula C_nH_{2n-2}. What is the value of n?

28. 29.2% (w/W) HCl stock solution has density of 1.25g mL^{-1}. The molecular weight of HCl is $36.5\,\text{g mol}^{-1}$. The volume (mL) of stock solution required to prepare a 200 mL solution 0.4 M HCl is

29. Ionisation potential and electron affinity of fluorine are 17.42 eV and 3.48 eV respectively. Determine the electronegativity of fluorine on Mulliken scale.

30. n-factor of $K_4[Fe(CN)_6]$ in the given reaction is -

$K_4[Fe(CN)_6] \xrightarrow{[O]} Fe^{3+} + NO_3^- + CO_2$

31. A rubber balloon weighing 10g is 40cm in diameter when inflated with He(g) at 760torr and $27^0 C$. Assuming density of air to be 1.2g/L under these conditions, the weight that the balloon can lift is

32. The number of waves made by a Bohr electron in H-atom for one complete revolution in its 3rd orbit are -

SECTION – III
(COMPREHENSIN TYPE)

This section contains 2 Paragraphs. Based on each paragraph, there are 2 questions. Each question has 4 options (A), (B), (C) and (D) for its answer, out of which **ONLY ONE** option can be correct.

Marking scheme: +3 for correct answer, 0 if not attempted and -1 in all other cases.

Paragraph for Question Nos. 33 to 34

Hydrogen spectrum is emission spectrum. It contains different spectral series. Wave number of spectral lines obtained in H-spectrum can be given as -

$$\bar{v} = \frac{1}{\lambda} = R\left(\frac{1}{x^2} - \frac{1}{y^2}\right)$$

Here x = lower orbit in which electron reaches

y = higher orbit from which electron jumps into lower orbit x

$R = 1.1 \times 10^7\ m^{-1}$ (Rydberg's constant)

Rough Work

33. If wavelength of series limit of Lyman series for He^+ ion is x Å, then wavelength of series limit of balmar series for Be^{3+} ion is -

(A) x Å (B) $\frac{9x}{4}$ Å (C) $\frac{16}{9}$ xÅ (D) $\frac{1}{x}$Å

34. The emission spectra is observed by the consequence of transition of electron from higher energy state to ground state of He^+ ion. Six different photons are observed during the emission spectra, then what will be minimum wavelength during transition ?

(A) $\frac{4}{27R_H}$ (B) $\frac{4}{15R_H}$ (C) $\frac{15}{16R_H}$ (D) $\frac{16}{15R_H}$

Paragraph for Question Nos. 35 to 36

A crystaline hydrated salt on being rendered anhydrous loses 45.6% of its weight.

The percentage composition of anhydrous salt is : Al = 10.5%, K = 15.1%, S = 24.8% and oxygen = 49.6%

[Molar mass are : Al = 27, K = 39, S = 32].

Answering following question based on these information.

35. What is the empirical formula of the anhydrous salt -

(A) $K_2AlS_2O_7$ (B) $K_2Al_2S_2O_7$

(C) $KAlS_2O_8$ D) $K_3AlS_2O_{12}$

36. What is the empirical formula of the hydrated salt?

(A) $K_2AlS_2O_7.10H_2O$ (B) $K_2Al_2S_2O_7.16H_2O$

(C) $K_3AlS_2O_{12}.8H_2O$ (D) $KAlS_2O_8.12H_2O$

Rough Work

MATHEMATICS MAX.MARKS: 60

SECTION- I
(Maximum Marks : 24)

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Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).
Negative Marks: -2 In all other cases.

37. The number of ordered pairs (m, n); where m, n $\in \{1,2,3,...,100\}$ such that $7^m + 7^n$ is divisible by 5 is

A) 1250 B) 2000 C) 2500 D) 5000

38. In the quadratic equation $A\left(\sqrt{3}-\sqrt{2}\right)x^2 + \dfrac{B}{\left(\sqrt{3}+\sqrt{2}\right)}x + C = 0$ with α, β as its roots, if

$A = \left(49+20\sqrt{6}\right)^{1/4}, B =$ sum of the infinite G.P as $8\sqrt{3} + \dfrac{8\sqrt{6}}{\sqrt{3}} + \dfrac{16}{\sqrt{3}} + ...\infty$ and $|\alpha - \beta| = \left(6\sqrt{6}\right)^k$,

where $k = \log_6 10 - 2\log_6 \sqrt{5} + \log_6 \sqrt{\left(\log_6 18 + \log_6 72\right)}$, then find the value of C.

A) 136 B) 164 C) 128 D) 64

39. Given that α, γ are roots of the equation $Ax^2 - 4x + 1 = 0$ and β, δ are roots of the equation $Bx^2 - 6x + 1 = 0$. If α, β, γ *and* δ are in H.P., then

A) $A = 5$ B) $A = 3$ C) $B = 8$ D) $B = -8$

40. $T_r = \dfrac{1}{r\sqrt{r+1} + (r+1)\sqrt{r}}$, then (here $r \in N$)

A) $T_r > T_{r+1}$ B) $T_r < T_{r+1}$ C) $\sum_{r=1}^{99} T_r = \dfrac{9}{10}$ D) $\sum_{r=1}^{n} T_r < 1$

Rough Work

41. If ${}^{100}C_6 + 4\,.{}^{100}C_7 + 6.\,{}^{100}C_8 + 4.\,{}^{100}C_9 + {}^{100}C_{10}$ has the value of equal to ${}^{x}C_y$; then the possible value (s) of $x+y$ can be

A) 112 B)114 C)196 D)198

42. The equation $\cos x \cos 6x = -1$

A) Has 50 solutions in $[0,100\pi]$ B)has 3 solution in $[0,3\pi]$

C) has even number of solutions in $(3\pi,13\pi)$ D) Has one solution in $\left[\frac{\pi}{2},\pi\right]$

SECTION - II
(Maximum Marks : 24)

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For each question, enter the correct numerical value (in decimal notation, truncated/rounded off to the **second decimal place**; e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30) designated to enter the answer.

Answer to each question will be evaluated according to the following marking scheme:

Full Marks: +3 If ONLY the correct numerical value is entered as answer.

Zero Marks: 0 In all other cases.

43. The number of positive integral solutions of $x_1x_2x_3x_4x_5 = 420$

44. Let $n \in N$ and S_n denotes the sum of all terms of a G.P, with first term $(1+x)^n$ (where $|x|<1$), common ratio be $\left(\frac{1+x}{2}\right)$ and the number of terms be $(n+1)$ then the sum of digits of the value of number of positive integral divisors of coefficient of x^{2015} in S_{2015} equals

45. The number of ways of distributing 3 identical physics books and 3 identical mathematics books among three students such that each student gets at least one book is 50 + K, where K is single digit number, then K is

46. If $\sin\alpha, \sin\beta, \sin\gamma$ are in A.P. and $\cos\alpha, \cos\beta, \cos\gamma$ are in G.P. then

$$\left|\frac{\cos^2\alpha + \cos^2\gamma - 4\cos\alpha\cos\gamma}{1-\sin\alpha\sin\gamma}\right| = \_\_\_\_\_\_\_\_\_\_$$

47. If $\alpha+\beta+\gamma=\pi$ and $\tan\left[\frac{\alpha+\beta-\gamma}{4}\right]\tan\left[\frac{\gamma+\alpha-\beta}{4}\right]\tan\left[\frac{\gamma+\beta-\alpha}{4}\right] = 1$ then the value of $1+\cos\alpha+\cos\beta+\cos\gamma$ is K - 1 where K is

48. Let $S_n = \frac{3}{4} + \frac{5}{36} + \frac{7}{144} + \frac{9}{400} + \ldots\ldots\ldots$ to 'n' terms. Then the digit in the 10's place of $\frac{1}{1-s_{40}}$ is

Rough Work

49. Let $P(x)=x^6-x^5-x^3-x^2-x$ and $\alpha,\beta,\gamma,\delta$ are the roots of the equation $x^4-x^3-x^2-1=0$ then $P(\alpha)+P(\beta)+P(\gamma)+P(\delta)=$

50. The number of solutions of the equation $(\sin x-1)^3+(\cos x-1)^3+(\sin x)^3=(2\sin x+\cos x-2)^3$ $in[0,2\pi]$

SECTION – III
(COMPREHENSIN TYPE)

This section contains 2Paragraphs. Based on each paragraph, there are 2 questions. Each question has 4 options (A), (B), (C) and (D) for its answer, out of which **ONLY ONE** option can be correct.

Marking scheme: +3 for correct answer, 0 if not attempted and -1 in all other cases.

Paragraph For Questions 51 and 52:

There are three cards in a bag each card having a different digits. The three players A,B,C play a game such that, in each round the cards are drawn from the bag, in order by A,B and C without replacement (after completion of every round the cards are put back in the bag). After two or more rounds, player B has sum total of numbers drawn 20, player A has sum total of numbers drawn 10 and player C has sum total of numbers 9. In the last round the player A received the largest number

51. The greatest number is
A) 7 B) 8 C) 6 D) 5

52. Total number of rounds played is
A) 5 B) 4 C) 3 D) 2

Paragraph For Questions 53 and 54:

Consider the equation $\left(x^2+x+1\right)^2-(m-3)\left(x^2+x+1\right)+m=0$ where m is a real parameter

53. The number of positive integral values of m for which the above equation has two distinct real roots is
A) 1 B) 2 C) 7 D) 6

54. The no. of positive integral values of m , $m\le 16$ for which the equation given above has 4 distinct real roots is
A) 1 B) 8 C) 7 D) 6

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Rough Work